Zero-Shot Vision Language Models for Energy-Theft Detection

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***Abstract*—Energy-theft detection on residential smart-meter data is a multi-class anomaly problem with severe class imbalance. Existing detectors consume engineered numeric features and emit a scalar anomaly score; off-the-shelf vision language models offer a parallel detector family but have not been evaluated on a representation that lets each see the same evidence. We define a *contextual pair*—a seven-day clean reference week paired with one query day across thirteen attack scenarios—and consume the same pair as 824 numeric features for four tree- and anomaly-based baselines (CatBoost, XGBoost, Random Forest, LOF) and as a single chart for four zero-shot vision LLMs (Gemini 3.1 Pro and Flash, Qwen3-VL-32B and 8B). On 97 never-seen consumers, the ML head is evaluated on the full 1.03 M-pair holdout and the LLM head on a 13 000-pair subset, sized via empirical sample-size stability curves so the inter-seed band (± 0.015 balanced accuracy) is narrower than the gap separating ML baselines from each other. CatBoost reaches AUC 0.986 and balanced accuracy 0.870; the best vision LLM reaches *[PENDING-LLM]*. Error analysis isolates a single dominant false-positive driver—per-consumer coefficient of variation—with tree-ensemble specificity dropping by $\sim$37 percentage points from the calmest to the noisiest consumer quartile. The contextual pair is a portable evaluation substrate, and high-variability consumers are the natural candidate population for an LLM-rationaled audit step.**

***Index Terms*—energy theft detection, smart meters, non-technical losses, vision language models, gradient boosting, anomaly detection, contextual prompting.**

I. Introduction

[PENDING — outline.md §2 P1: smart-meter rollout, NTL cost, scale of audit problem.]

[PENDING — outline.md §2 P2: established ML detection paradigms, what they share.]

[PENDING — outline.md §2 P3: contextual-pair representation, ML+VLM head-to-head.]

[PENDING — outline.md §2 P3.5: anchor numbers (CatBoost 0.986, Q4 spec 0.566, LLM = PENDING).]

[PENDING — outline.md §2 P4: utility audit-cost framing.]

Our contributions are:

1) **Contextual-pair representation.** A reproducible per-consumer-month construction (7-day reference + 1 query day across 13 attack scenarios) consumed identically as 824 numeric features and as a single chart for vision-LLM prompting. *[PENDING — full sentence.]*
2) **Head-to-head ML vs. vision-LLM evaluation on identical pairs.** Four tree/anomaly baselines on 1.03 M never-seen-consumer pairs; four off-the-shelf vision LLMs zero-shot on a stability-calibrated subset of the same rows. *[PENDING-LLM.]*
3) **Stability-calibrated VLM evaluation protocol.** Empirical sample-size stability curves (Section III-E1) justify a tractable subset for vision-LLM benchmarking that would otherwise require multi-week-per-model GPU time. *[PENDING — full sentence.]*
4) **High-variability consumers as a post-hoc-discovered failure axis.** Error analysis on the held-out predictions surfaces per-consumer CV as the dominant false-positive driver; tree-ensemble specificity drops $\sim$37 pp from the calmest to the noisiest CV quartile (Section IV-C). *[PENDING — full sentence.]*

II. Background and Problem Setting

A. Notation and Problem Definition

[PENDING — see outline.md §3.1.]

B. Attack Catalogue

[PENDING — see outline.md §3.2.]

III. Methods

A. Dataset and Consumer Split

[PENDING.]

B. Contextual-Pair Construction

[PENDING.]

C. ML Baselines

[PENDING.]

D. Vision Language Models and Prompting

[PENDING-LLM — system prompt will be lifted from pipeline/llm/prompts/classify.txt.]

E. Evaluation Protocol

[PENDING.]

1) Sample-Size Calibration via Stability Curves: [PENDING — see outline.md §4.5.1.]

IV. Results

A. Head-to-Head: Tree Ensembles vs. Vision LLMs

[PARTIAL — ML rows ready; LLM rows PENDING.]

B. Per-Attack Identification

[PENDING — Fig. 4 bars + takeaway.]

C. Error Analysis: A Discovered Failure Axis

[PENDING — see outline.md §5.3.]

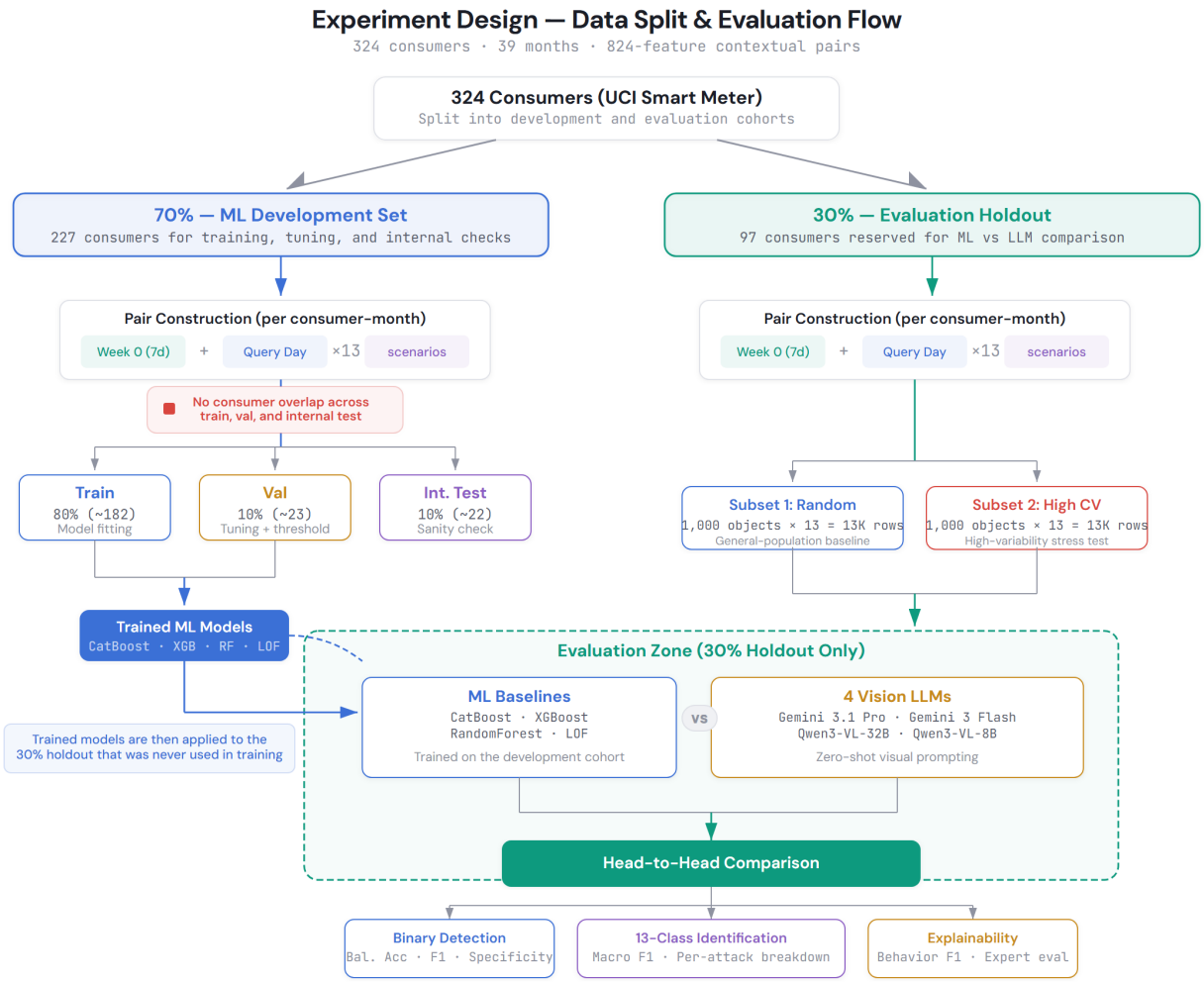

Fig. 1. Experiment design. The 324-consumer UCI cohort splits into a 70% ML development set (227 consumers, with an 80/10/10 inner train/val/internal-test split) and a 30% evaluation holdout (97 consumers, never seen during ML training). For both branches, each consumer-month produces pairs of (a) a 7-day clean reference week (week 0) and (b) one query day from weeks 1–3 across 13 attack scenarios. Trained ML models are then applied to the 30% holdout, where four off-the-shelf vision LLMs evaluate the same rows zero-shot. The Random and High-CV subsets sample identical rows for ML and LLM heads.

TABLE I
BINARY DETECTION ON THE 97-CONSUMER HOLDOUT (1.03 M PAIRS, ∼12:1 THEFT:CLEAN). THRESHOLD TUNED ON VAL FOR MAX F1.

Model	AUC	Bal. Acc	F1	Recall	Spec.	Prec.
CatBoost	0.986	0.860	0.982	0.987	0.733	0.978
XGBoost	0.986	0.870	0.982	0.984	0.755	0.980
RandomForest	0.977	0.822	0.977	0.982	0.663	0.972
LOF (unsup.)	0.628	0.549	0.282	0.165	0.933	0.967
Gemini 3.1 Pro	—	—	—	—	—	—
Gemini 3 Flash	—	—	—	—	—	—
Qwen3-VL-32B	—	—	—	—	—	—
Qwen3-VL-8B	—	—	—	—	—	—

TABLE II
SPECIFICITY BY PER-CONSUMER CV QUARTILE (CLEAN ROWS ONLY).

Quartile	CatBoost	XGBoost	RF	LOF
Q1 (low CV)	0.938	0.950	0.904	0.990
Q2	0.692	0.705	0.658	0.978
Q3	0.727	0.770	0.638	0.941
Q4 (high CV)	**0.566**	**0.588**	**0.442**	0.822
Overall	0.733	0.755	0.663	0.933

V. DISCUSSION

[PENDING.]

VI. RELATED WORK

[PENDING.]

VII. CONCLUSION

[PENDING.]

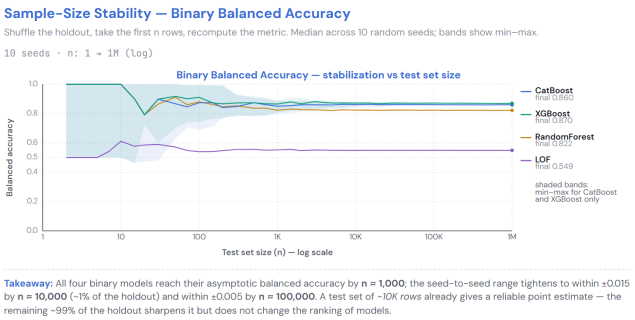

Fig. 2. Binary balanced-accuracy stability. We shuffle the 1.03 M-pair holdout, take the first n rows, recompute the metric, and repeat across ten seeds; the median is plotted with min–max bands for the two top models. All four ML detectors reach their asymptotic balanced accuracy by $n \approx 1\,000$; the inter-seed band tightens to ± 0.015 by $n \approx 10\,000$. The vision LLMs are evaluated at $n = 13\,000$.

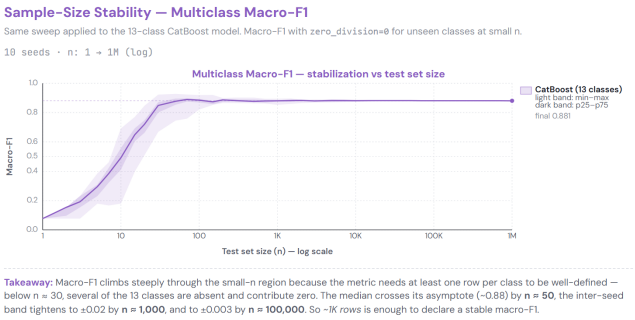

Fig. 3. 13-class macro-F1 stability. Median crosses the asymptote (~ 0.88) by $n \approx 50$; the inter-seed band tightens to ± 0.02 by $n \approx 1\,000$ and ± 0.003 by $n \approx 100\,000$.

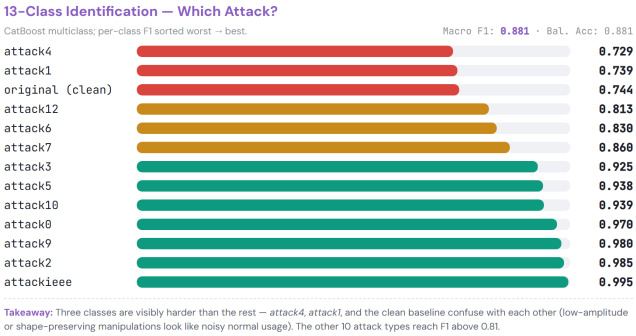

Fig. 4. 13-class identification — per-class F1 sorted worst-first. CatBoost reaches macro-F1 = 0.881 overall; three classes (*attack4*, *attack1*, original) confuse with each other — low-amplitude or shape-preserving manipulations resemble noisy normal usage. The other ten attack types reach F1 above 0.81.